

Burns: Electrical/Lightning Injury

Guideline Title: Burns: Electrical/Lightning Injury

Guideline Number: GUID-3203-T005

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Revision dates: 6/8/2021	Developed by: Air Care Team
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CVI. **PURPOSE:** To maintain adequate tissue perfusion, obtain an accurate mechanism of injury, and transport the patient to appropriate facility with burn treatment capabilities. Electrical injury may result from contact with alternating current, direct current or lightning. Small surface wounds may be associated with devastating internal injuries.

CVII. **DEFINITIONS:**

A. TBSA – Total body surface area

CVIII. **DEPARTMENT GUIDELINE:**

- A. Safe extrication from electrical source by experienced rescue personnel.
- B. Perform a general trauma assessment and assess and manage airway, breathing, and circulation.
 - i. Administer 100% oxygen per non-rebreather mask.
 - ii. If necessary, secure an advanced airway *per GUID-G008 Rapid Sequence Induction for Endotracheal Intubation and GUID-PR020 Endotracheal Intubation* and ventilate with a transport ventilator *per GUID-PR024 Ventilation with a Mechanical Ventilator*.
- C. Assess for mechanism of injury and any trauma circumstances surrounding the injury.
 - i. What kind of electricity was involved – high voltage/low voltage, AC/DC?
 - ii. What was the duration of contact?
 - iii. Was the patient thrown or did he or she fall?
 - iv. Was there a loss of consciousness?
 - v. Was CPR administered at the scene?
- D. Stop the burning process. Remove all clothing and metal.
- E. All patients at risk for spinal injury need to be immobilized.
- F. Assess for entry and exit wounds.
- G. Assess burned extremities for distal pulses, motor, and sensory function.
- H. Assess for signs of impending or present shock and treat if present.
- I. Assume and treat for closed head injury in any unconscious but otherwise stable patient *per GUID-T014 - Traumatic Brain Injury*.
- J. Apply cardiac monitor, monitor, and treat dysrhythmias *per GUID-C001-C010 - Cardiac Guidelines*.
 - i. Consider obtaining 12 lead EKG

- ii. In multiple casualty electrical injuries, any victim in asystole or unstable cardiac status should be resuscitated first (Reverse triage principle).
- K. Establish vascular access.
- L. Infuse crystalloid IV fluids (warm if available) sufficient to maintain SBP of >90 mmHg in adult patients and patients older than 10 years and >70 plus twice the age in pediatric patients <10 years old.
- M. Keep patient warm. Apply blankets.
- N. Perform a detailed motor/sensory neurological examination & document changes. There is a greater possibility of nerve damage & compartment syndrome with minimal cutaneous injury.
- O. Treat pain and/or anxiety per *GUID-G005 – Analgesia and Sedation Management* and *GUID-3203-G006 Anxiety/Combative patient Management*.
- P. If there is evidence of myoglobinuria (elevated CK, decreased urine output), the medical transport team should **consider**:
 - i. Adult: **Sodium Bicarbonate** 8.4% (50 mEq) IV x 1.
 - ii. Pediatric: **Sodium Bicarbonate** 8.4% 1 mEq/kg IV x 1, max dose 50 mEq.
 - iii. Adult patients with high voltage electrical injuries with evidence of myoglobinuria should have a urine output of 1-1.5ml/kg/h (approx. 75-100ml/h) until urine clears.
- Q. All electrical burns, including lightning injury should be transported to a burn center in the absence of significant non-burn related trauma.
- R. Burn centers in the State of Florida include Orlando Regional Medical Center, Tampa General Hospital, Shands Health Care University of Florida – Gainesville, Jackson Memorial Hospital - Miami, Kendall Regional Medical Center -Miami, FL, and Blake Medical Center - Bradenton.
- S. High voltage electric injury (1,000 volts or greater) is also considered a trauma alert for Orlando Regional Medical Center.

CIX. **DOCUMENTATION:**

- A. EMS Charts

CX. **REFERENCES:**

- A. American Burn Association. (2016). *Advanced Burn Life Support Course Provider Manual*.